

Changyu (Thomason) Zhao

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EDUCATION

University of Wisconsin-Madison

Bachelor of Science in Computer Science (Honors) and Math

Cumulative GPA: **3.97/4.0**, Dean's List 2021 - 2023

Sep. 2021 – May 2025

Madison, Wisconsin

EXPERIENCES

UW-Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – Present

Madison, Wisconsin

- Managed security incidents, including responding to alerts, investigating security breaches, and taking corrective action.
- Used **Elastic** and **Palo Alto Firewall** to **monitor and analyze campus network traffic**, identify potential threats and vulnerabilities, and notify security alerts to relevant users or departments.
- Conducted **Identity and Access Management (IAM)** procedures to manage access for full-time and student employee accounts.
- Collaborated with team members to **process daily threat intelligence**, fostering a **collaborative environment** to address security concerns and mitigate potential threats on campus.

ASU SEFCOM/CDF Security Research Apprenticeship

Research Assistant

May. 2023 – Aug. 2023 & May. 2024 – Sep. 2024

Tempe, Arizona

- Conceptualized, designed, and implemented **LibHopper**, a Python-based security research tool designed to **automate the discovery of exploitation primitives** within arbitrary user libraries.
- Leveraged automated processes to **pipeline the evaluation of security implications** related to memory corruption vulnerabilities in library internal structures, enhancing the capabilities of security researchers and developers.
- Employed a multifaceted methodology involving **customizing debugger scripts** and **symbolic execution techniques** to conduct comprehensive **exploitability analysis**.
- Identified and cataloged **exploitable primitives** within library internal states, derived from meticulous analysis of memory corruption scenarios and symbolic execution findings on the 30 most widely-used C standard libraries.

Tencent Security Xuanwu Lab

Security Research Intern

Aug. 2023

Beijing, China

- Developed **LibIHT**, a cross-platform (Linux and Windows), **hardware-assisted dynamic control flow analysis** tool in **C and C++** that leverages Intel's **Last Branch Record (LBR)** and **Branch Trace Store (BTS)** hardware tracing features for enhanced system evaluation.
- Designed and implemented **kernel modules** and **device drivers** to enable tracing features for Linux and Windows.
- Architected and developed a user-mode analysis library that leverages collected trace information to **accurately reconstruct the control flow graph (CFG)** of executed programs. The library incorporated features for parsing, analysis, and visualization.

PROJECTS

Router automated vulnerability discovery | *Reverse Engineering, Firmware, Python*

May 2022 – Aug. 2022

- Discovered and reported multiple previously **unknown vulnerabilities** (CVE-2018-16334, CVE-2018-5767, etc.) in the Tenda AC18 router firmware.
- Developed an **automated IDA Pro plugin** to systematically detect **stack overflow and command injection vulnerabilities** in the decompiled router firmware.
- Leveraged **QEMU** to **rehost the ARM-based router firmware** on an x86-64 machine, enabling **in-depth debugging and analysis**.
- Demonstrated the ability to **achieve Remote Code Execution (RCE)** by exploiting the identified vulnerabilities.

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Team member

Sep. 2021 – Present

China

- Achieved a notable accomplishment by securing a rank 12/1282 worldwide in the DEFCON 30 CTF Finals (2022), and 7/1828 worldwide in the DEFCON 31 CTF Finals (2023).
- Participated in over 40 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.

pwn.college

Independent Learning

Dec. 2021 – Jul. 2022

Tempe, Arizona

- Attained the Yellow Belt and Blue Belt levels by completing **all** challenges in Dojo instance, currently ranked 43/2500+.
- Acquired comprehensive skills in various areas of **software and system security**, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Proficient in reverse engineering, have reverse experience in large-scale projects. **Familiar** with control flow tracking, data flow tracking, symbolic execution and the use of binary analysis tools.

Programming Languages: C/C++, Python, Java, Assembly Language, JavaScript, HTML/CSS, L^AT_EX

Binary Security Tools: IDA Pro/Ghidra, GDB/Windbg, Angr, Qemu, Frida, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux

Soft Skills: Time/Project Management, Collaboration, Conflict Resolution, Active Listening